

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

Find the quotient. Write it in rectangular form, using exact values. (3 points)

2. $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4. $x^4 + 8 + 8i\sqrt{3} = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = -2$

YES / NO

b) $z = -1 + i$

YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

Every line shown. Point splits are an estimate of partial credit (halves are possible).

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

1. Multiply the r 's, add the θ 's (common denominator first).

$$r = \sqrt{2} \cdot \sqrt{2} = 2 \quad \theta = \frac{5\pi}{4} + \frac{11\pi}{12} = \frac{15\pi}{12} + \frac{11\pi}{12} = \frac{26\pi}{12} = \frac{13\pi}{6} \xrightarrow{-2\pi} \frac{\pi}{6}$$

$$2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} + i \left(\frac{1}{2} \right) \right) = \boxed{\sqrt{3} + i}$$

Trap: given in radians, stay in radians. $2\pi = \frac{12\pi}{6}$.

Find the quotient. Write it in rectangular form, using exact values. (3 points)

Problem: $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

Top: $r = \sqrt{(\sqrt{2})^2 + (\sqrt{6})^2} = \sqrt{8} = 2\sqrt{2} \quad \tan \theta = \frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3} \Rightarrow \text{QI}, 60^\circ$

Bottom: $r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2 \quad \tan \theta = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QIII} \Rightarrow 210^\circ$

$$\frac{2\sqrt{2} \text{cis } 60^\circ}{2 \text{cis } 210^\circ} = \sqrt{2} \text{cis}(-150^\circ) \xrightarrow{+360^\circ} \sqrt{2} \text{cis } 210^\circ = \sqrt{2} \left(-\frac{\sqrt{3}}{2} + i \left(-\frac{1}{2} \right) \right) = \boxed{-\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i}$$

Trap: both parts of the bottom negative means QIII: 210° , not 30° .

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

3. De Moivre: raise r to the 4th, multiply the angle by 4.

$$(\sqrt{2})^4 = 4 \quad 4 \cdot \frac{2\pi}{3} = \frac{8\pi}{3} \xrightarrow{-2\pi} \frac{8\pi}{3} - \frac{6\pi}{3} = \frac{2\pi}{3}$$

$$4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = 4 \left(-\frac{1}{2} + i \left(\frac{\sqrt{3}}{2} \right) \right) = \boxed{-2 + 2\sqrt{3}i}$$

Trap: $(\sqrt{2})^4 = 4$, not $4\sqrt{2}$. $\frac{2\pi}{3}$ is QII: cos negative, sin positive.

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

Problem: $x^4 + 8 + 8i\sqrt{3} = 0$

4. Isolate x^4 first (both signs flip), convert, then the n th root theorem in radians.

$$\begin{aligned} x^4 &= -8 - 8i\sqrt{3}: \quad r = \sqrt{(-8)^2 + (-8\sqrt{3})^2} = \sqrt{64 + 192} = \sqrt{256} = 16 \\ \tan \theta &= \frac{-8\sqrt{3}}{-8} = \sqrt{3} \Rightarrow \text{ref } \frac{\pi}{3}, \text{ QIII} \Rightarrow \frac{4\pi}{3} \\ x &= (16 \text{cis } \frac{4\pi}{3})^{1/4} = \sqrt[4]{16} \text{cis } \frac{4\pi/3}{4} = 2 \text{cis } \frac{\pi}{3} + \frac{2\pi}{4} = +\frac{\pi}{2} \\ 2 \text{cis } \frac{\pi}{3} &= 2\left(\frac{1}{2}\right) + 2i\left(\frac{\sqrt{3}}{2}\right) = 1 + \sqrt{3}i \quad 2 \text{cis } \frac{5\pi}{6} = 2\left(-\frac{\sqrt{3}}{2}\right) + 2i\left(\frac{1}{2}\right) = -\sqrt{3} + i \\ 2 \text{cis } \frac{4\pi}{3} &= 2\left(-\frac{1}{2}\right) + 2i\left(-\frac{\sqrt{3}}{2}\right) = -1 - \sqrt{3}i \quad 2 \text{cis } \frac{11\pi}{6} = 2\left(\frac{\sqrt{3}}{2}\right) + 2i\left(-\frac{1}{2}\right) = \sqrt{3} - i \end{aligned}$$

$$x = 1 + \sqrt{3}i, -\sqrt{3} + i, -1 - \sqrt{3}i, \sqrt{3} - i$$

Trap: moving $8 + 8i\sqrt{3}$ across flips both signs: $x^4 = -8 - 8i\sqrt{3}$ is QIII ($\frac{4\pi}{3}$), not QI. Check: $(2 \text{cis } \frac{\pi}{3})^4 = 16 \text{cis } \frac{4\pi}{3}$.

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

Problem: a) $z = -2$

YES / NO

b) $z = -1 + i$

YES / NO

5. Keep computing until an absolute value goes over 2 (then NO) or the values start repeating (then YES).

a) $z = -2$: $|-2| = 2$ (equal to 2 does not "exceed" 2, keep going)

$$z^2 + z = 4 - 2 = 2, \quad |2| = 2$$

$(2)^2 + (-2) = 2$ again: stuck at 2 forever, never over 2 YES

b) $z = -1 + i$: $|z| = \sqrt{2} \approx 1.4$

$$z^2 + z = (1 - 2i + i^2) + (-1 + i) = -2i - 1 + i = -1 - i, \quad |-1 - i| = \sqrt{2}$$

$$(z^2 + z)^2 + z = (-1 - i)^2 + (-1 + i) = (1 + 2i + i^2) - 1 + i = 2i - 1 + i = -1 + 3i$$

$$|-1 + 3i| = \sqrt{1 + 9} = \sqrt{10} \approx 3.2 > 2 \quad \text{NO}$$

Trap: b) passes the first two checks and fails on the third. Don't stop early and call it YES.